

Advanced ASIC Design

Exercise no 1: SPYGLASS LINT

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3. SpyGlass lint

3.1 initial_rtl stage

3.1.1 First violations

The following 3 violations were reported by all lints (Goals lint/lint_rtl, lint/lint_rtl_enhanced, lint/lint_turbo_rtl, lint/lint_functional_rtl):

1. STX_VE_274 (1) : The Param assignment list must be complete.

Message: *Param assignment list is incomplete*

Location: RTL/s7_display.v:22

Reason: trailing comma in parameter list

Solution: remove trailing comma (in line 21)

Comment: The list is not incomplete - this is just a syntax error.

2. STX_VE_479 (2) : A Verilog-2005/SystemVerilog/SystemVerilog-2009 construct cannot be used without specifying corresponding 'set_option'.

1. **Message:** *Syntax error near (;). Please use 'set_option enableSV yes' to support SystemVerilog construct (Null statement)*

Location: RTL/bcd_mux.v:24

Reason: duplicate semicolon creates a null statement

Solution: remove one of the semicolons (in line 24)

2. **Message:** *Syntax error near (function). Please use 'set_option enableSV yes' to support SystemVerilog construct (Function declaration outside module scope)*

Location: RTL/bcd_mux.v:50

Reason: function clogb2 is declared outside module

Solution: move the function to bcd_mux module scope, by moving endmodule from line 48 to the end of file

3.1.2 Second iteration

After fixing the violations listed above, all goals reported the following 2 violations:

1. STX_VE_606 (2) : Only those identifiers can be used that are declared in the current scope.

Message: *Identifier (bcd) not declared in current scope*

Locations: RTL/decoder.v:9, RTL/decoder.v:11

Reason: typo in signal name

Solution: replace both instances of bcd with i_bcd

3.1.3 Third iteration

After fixing the violation listed above, all goals reported the following 4 violations:

1. **STX_VE_361 (4) : A procedural assignment statement cannot drive a net other than reg type.**

Message: *Procedural assignment statement cannot drive a net : r_display_count*

Locations: RTL/bcd_mux.v:33, RTL/bcd_mux.v:36, RTL/bcd_mux.v:37, RTL/bcd_mux.v:38

Reason: r_display_count is declared as wire but used like reg

Solution: change wire to reg in r_display_count declaration (in line 29)

3.1.4 Fourth iteration

After fixing the violation listed above, the following violations were reported:

1. **STARCO5-2.11.3.1 (1) : Sequential and combinational parts of an FSM description should be separated**

Message: *Combinational and sequential parts of an FSM 'bcd_mux.r_display_count' described in same always block*

Goals: lint/lint_rtl, lint/lint_turbo_rtl

Location: RTL/bcd_mux.v:37

Reason: SpyGlass considers this an FSM and is unhappy about describing combinational logic in assignment

Solution: calculate the next value of r_display_count as a separate signal

2. **W71 (1) : A case statement(or selected signal assignment) does not have a default or OTHERS clause**

Message: *Case statement does not have a default clause and is not preceded by assignment of target signal in combinational block[Hierarchy: 's7_display:decoder_i@decoder']*

Goals: lint/lint_rtl, lint/lint_rtl_enhanced, lint/lint_turbo_rtl

Location: RTL/decoder.v:11

Reason: segments may not have assigned value and becomes an inferred latch

Solution: add default case that assigns 7'b1111111 (all off) to segments or make this assignment in the same always, before case

Comment: the case is also missing a case for i_bcd == 5

3. **W110 (1) : An instance port connection has incompatible width compared to the port definition**

Message: *Incompatible width for port 'o_segments'(width 7 in module 'decoder') on instance 'decoder_i'(actual width 1) [Hierarchy: 's7_display']*

Goals: lint/lint_rtl, lint/lint_rtl_enhanced, lint/lint_turbo_rtl

Location: RTL/s7_display.v:36

Reason: segments is not declared and because the design does not use `default_nettype none, a wire (of width 1) is created by default

Solution: declare `segments` net (type `wire`) of width 7 (`[6:0]`, same as `o_segments`)

4. **W486 (1) : Shift overflow - some bits may be lost**

Message: *Rhs width '4' with shift (Expr: '({{ (DISPLAYS_NUM - 1){ 1'b0 } }, 1'b1 } « r_display_count)') is more than lhs width '1' (Expr: 'bcd_sel'), this may cause overflow [Hierarchy: ':s7_display:bcd_mux_i@bcd_mux']*

Goals: lint/lint_rtl, lint/lint_rtl_enhanced, lint/lint_turbo_rtl

Location: RTL/bcd_mux.v:46

Reason: `bcd_sel` is not declared (a typo, should be `o_bcd_sel`) and because the design does not use ``default_nettype none`, a wire (of width 1) is created by default

Solution: change `bcd_sel` (in line 46) to `o_bcd_sel`

5. **InferLatch (1) : Latch inferred**

Message: *Latch inferred for signal 'segments[6:0]' in module 'decoder'*

Goals: lint/lint_rtl, lint/lint_rtl_enhanced, lint/lint_turbo_rtl

Location: RTL/decoder.v:20

Reason: see point 2. on this list (missing default)

Solution: see point 2. on this list (missing default)

6. **W336 (3) : Blocking assignment should not be used in a sequential block (may lead to shoot through)**

1. **Message:** *Blocking assignment 'r_sel_counter = (r_sel_counter + 1);' used inside a 'FlipFlop' inferred sequential block*

Goals: lint/lint_rtl, lint/lint_rtl_enhanced, lint/lint_turbo_rtl

Location: RTL/bcd_mux.v:24

Reason: a blocking assignment (`=`) is used when assigning to a flip-flop instead of a non-blocking one (`<=`)

Solution: change the assignment to non-blocking (`<=`)

2. **Message:** *Blocking assignment 'r_sel_counter = 0;' used inside a 'FlipFlop' inferred sequential block*

Goals: lint/lint_rtl, lint/lint_rtl_enhanced

Locations: RTL/bcd_mux.v:20, RTL/bcd_mux.v:23

Reason: blocking assignments (`=`) are used when assigning to a flip-flop instead of non-blocking ones (`<=`)

Solution: change the assignments to non-blocking (`<=`)

7. **W528 (1) : A signal or variable is set but never read**

Message: *Variable 'bcd_sel' set but not read.[Hierarchy: ':s7_display:bcd_mux_i@bcd_mux']*

Goals: lint/lint_rtl, lint/lint_turbo_rtl

Location: RTL/bcd_mux.v:46

Reason: `bcd_sel` is not declared (a typo, should be `o_bcd_sel`) and because the design does not use ``default_nettype none`, a wire (of width 1) is created by default, but other places expect the correct name

Solution: change `bcd_sel` (in line 46) to `o_bcd_sel`

8. AutomaticFuncTask-ML (1) : Use automatic functions/tasks in modules and interfaces.

Message: *Function 'clogb2' should be declared as automatic [Hierarchy: 's7_display:bcd_mux_i@bcd_mux']*

Goals: lint/lint_rtl_enhanced

Location: RTL/bcd_mux.v:48

Reason: function `clogb2` is not declared as automatic and is static by default

Solution: make the function automatic (`function automatic integer clogb2;`)

9. STARC-2.3.4.3 (2) : A flip-flop should have an asynchronous set or an asynchronous reset

1. **Message:** *Flip-flop 'r_display_count[1:0]' has neither asynchronous set nor asynchronous reset. [Hierarchy: 's7_display.bcd_mux_i']*

Goals: lint/lint_rtl_enhanced

Location: RTL/bcd_mux.v:38

Reason: the flip-flop has synchronous reset

Solution: add `negedge i_rst` to the sensitivity list to make reset asynchronous

2. **Message:** *Flip-flop 'r_sel_counter[3:0]' has neither asynchronous set nor asynchronous reset. [Hierarchy: 's7_display.bcd_mux_i']*

Goals: lint/lint_rtl_enhanced

Location: RTL/bcd_mux.v:24

Reason: the flip-flop has synchronous reset

Solution: add `negedge i_rst` to the sensitivity list to make reset asynchronous

10. UndrivenOutPort-ML (1) : Undriven but loaded output port of a module detected

Message: *Detected undriven output port o_bcd_sel[3:0]*

Goals: lint/lint_rtl_enhanced

Location: RTL/bcd_mux.v:12

Reason: `o_bcd_sel` is undriven, because of a typo in assignment in line 46

Solution: change `bcd_sel` in line 46 to `o_bcd_sel`

11. W164b (2) : LHS width is greater than RHS width of assignment (Extension)

1. **Message:** *LHS: 'o_segments' width 7 is greater than RHS: 'segments' width 1 in assignment [Hierarchy: 's7_display']*

Goals: lint/lint_rtl_enhanced

Location: RTL/s7_display.v:39

Reason: `segments` is not declared and because the design does not use ``default_nettype none`, a wire (of width 1) is created by default

Solution: declare `segments` net (type `wire`) of width 7 (`[6:0]`), same as

o_segments)

2. **Message:** *LHS: 'bcd_out' width 4 is greater than RHS: 'i_bcd_data[(4 * ((DISPLAYS_NUM - r_display_count) - 1)) +:3]' width 3 in assignment [Hierarchy: ':s7_display:bcd_mux_i@bcd_mux']*

Goals: lint/lint_rtl_enhanced

Location: RTL/bcd_mux.v:42

Reason: part select is +:3, but 4 bytes should be selected

Solution: change part select to +: 4

12. **W241 (2) : Output is never set**

Message: *Output 'o_bcd_sel[3:0]' is never set.[Hierarchy: ':s7_display:bcd_mux_i@bcd_mux']*

Goals: lint/lint_rtl_enhanced

Location: RTL/bcd_mux.v:12

Reason: o_bcd_sel is undriven, because of a typo in assignment in line 46

Solution: change bcd_sel in line 46 to o_bcd_sel

13. **Av_init01 (1) : Reports initial setup issues of the design.**

Message: *Could not find clocks for all the flops. Please add clock SGDC constraint to the design*

Goals: lint/lint_functional_rtl

Location: RTL/s7_display.v:1

Reason: constraints file was not included in project

Solution: add RTL/s7.sgdc to project (read_file -type sgdc RTL/s7.sgdc)

3.1.5 Fifth iteration

No violations reported.

3.2 rtl_handoff stage

No violations reported.

4. Simulation

When “5” should be displayed, `segments` do not form a valid digit (is all ones).

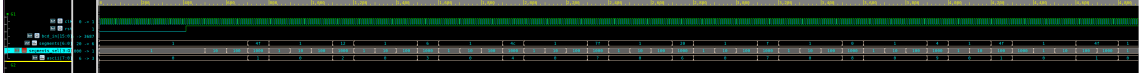


Figure 4.1: Simulation waveform showing segments 0x7F not decoding to any digit, where 5 was expected. The PDF can be zoomed in to show waveform in more details.

This is because in `decoder.v` the `case (i_bcd)` has no case for 5 as already found when analysing **violation 2 in fourth iteration**. The missing case is

```
5 : segments = 7'b0100100;
```

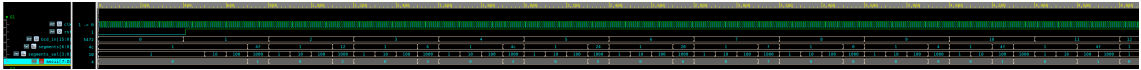


Figure 4.2: Waveform after fixing with digit 5 (segments 0x24) where it should be.

5. Additional “guideline” rules

After enabling a few rules labelled as “guideline”, the following violations were reported:

1. **ClkName (2) : Clock name does not follow the naming convention.**

Message: *Clock signal name ‘i_clk’ should be clk or start with clk_*

Locations: RTL/s7_display.v:26, RTL/bcd_mux.v:19

Solution: rename clock signal to clk (remember about .sgdc)

2. **ParamName (4) : Parameter/generic name does not follow the naming convention**

1. **Message:** *Parameter name DISPLAYS_NUM does not follow naming convention*

Locations: RTL/s7_display.v:3, RTL/bcd_mux.v:3

Reason: parameter name must not be longer than 8 characters

Solution: rename parameter to DISP_NUM

2. **Message:** *Parameter name MULTIPLEX_CLK_COUNT does not follow naming convention*

Locations: RTL/s7_display.v:4, RTL/bcd_mux.v:4

Reason: parameter name must not be longer than 8 characters

Solution: rename parameter to CC_PER_D

3. **RegOutName (2) : Register output signal does not follow naming convention.**

1. **Message:** *Register output name ‘r_sel_counter’ should end with _r*

Location: RTL/bcd_mux.v:24

Solution: rename to sel_counter_r

2. **Message:** *Register output name ‘r_display_count’ should end with _r*

Location: RTL/bcd_mux.v:37

Solution: rename to display_count_r

4. **ResetName (1) : Reset name does not follow the naming convention.**

Message: *Reset signal name ‘i_rst’ should start with rst*

Location: RTL/bcd_mux.v:36

Solution: rename to rst (remember about .sgdc)

5. **SepTFMacro-ML (1) : Task/functions/macros should be defined in separate file**

Message: *Define task/functions/macros in a separate file*

Location: RTL/bcd_mux.v:46

Solution: Move the function to a separate file (clogb2.v) and `include it

It is a good idea to specify and enforce code structuring and naming rules in a project to maintain internal consistency. However, there is no single naming/formatting standard that must be followed and style is a matter of opinion.

In this case, respecting the above rules, means breaking the existing code style (`i_*` for inputs, `o_*` for outputs and `r_*` for registers), because registers must use a suffix and clock and reset cannot have the same prefix as other inputs. Additionally, the *ParamName* rule (amongst other things) restricts the name length to 8 characters, because longer names “*reduce readability*”. Changing parameter names to adhere to this rule, reduces readability much more...

Personally, I prefer using linters and formatting tools to keep the code style mostly consistent, especially when working with others. When enabling such rules, the exact specifics should be thoroughly thought out to:

- not overspecify rules to the point where developers are constantly fighting or working them around,
- enforce style that is readable, not just arbitrary, and
- be strict enough to actually require a certain style.

In case of this project, only the *SepTFMacro-ML* rule makes sense and others should be disabled, or custom requirements should be defined instead. The activation of such rules is by no way necessary, as it affects only code style/structure.

6. Stopwatch

The stopwatch implementation required changing the number of displays to 6, which revealed another problem in `bcd_mux.v`. `r_display_count` has a range `[0, DISPLAYS_NUM]` (inclusive), but the upper bound should be exclusive (maximum allowed value of `DISPLAYS_NUM-1`). This could not be observed with `DISPLAYS_NUM` equal to a power of 2, such as 4 previously, because of `r_display_count` width. Fixing this required changing `r_display_count == DISPLAYS_NUM` comparison to one with `DISPLAYS_NUM-1`.

While writing the stopwatch, the initial version did have synchronous reset, which caused violation shown on Fig. 6.1.

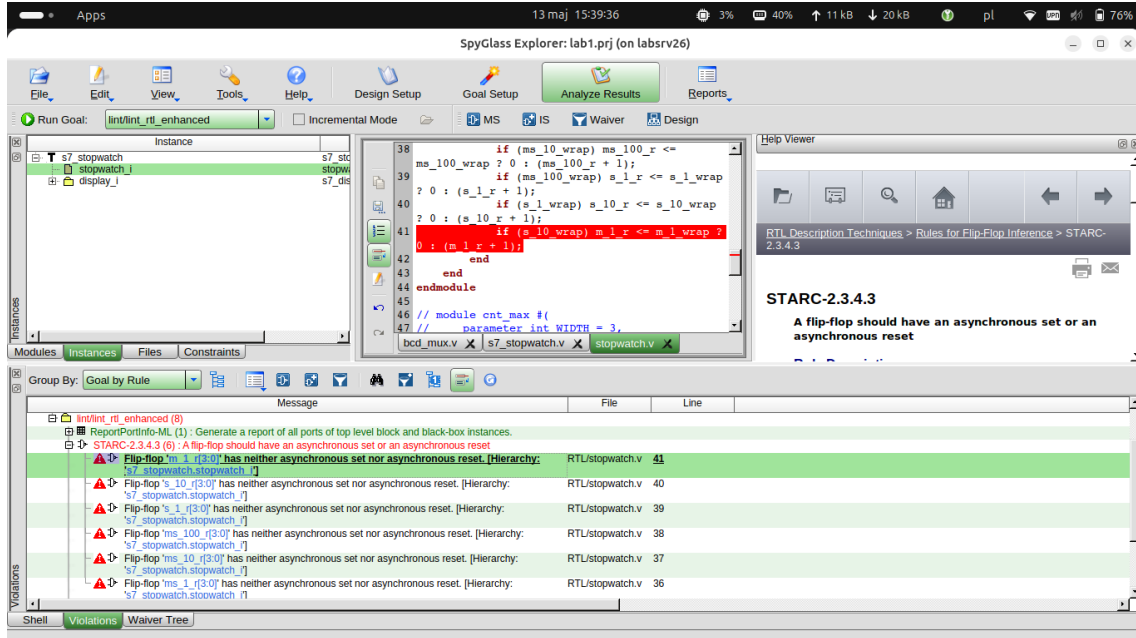


Figure 6.1: Violations reported for initial version of stopwatch.

In the final version, goals `lint/lint_rtl` and `lint/lint_turbo_rtl` report violation in `RTL/stopwatch.v:51`:

FlopEConst (1) : Flip-flop enable pin is permanently disabled or enabled
 Enable pin EN on Flop `s7_stopwatch.stopwatch_i.ms_1_r[0]` (master RTL_FDCE) is always enabled (tied high)(connected to `s7_stopwatch.stopwatch_i.sub_ms_wrap`)

This violation occurs, because when `CYCLES_PER_MS` is 1, signal `sub_ms_wrap` is assigned constant 1, by design. This is expected and for greater values of

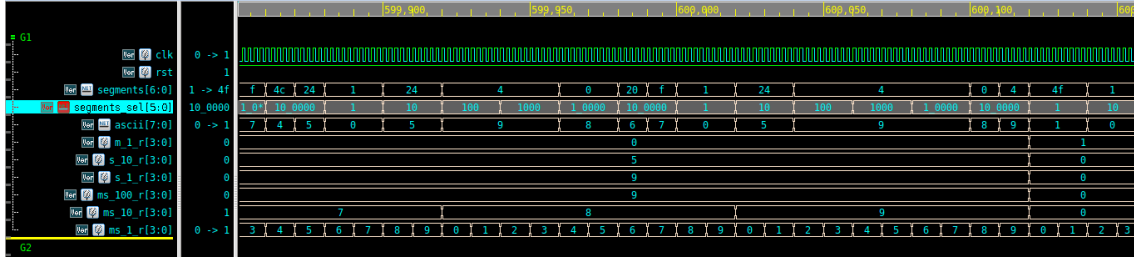


Figure 6.2: Waveform from stopwatch simulation showing the counter approaching 0:59.999 and changing to 1:00.000. Simulation was performed with 5 kHz clock.

CYCLES_PER_MS there is no violation, as conditional generate follows a different path, that does not assign a constant.

Source code of the stopwatch module from RTL/stopwatch.v:

```

1 module stopwatch #(
2     parameter CYCLES_PER_MS = 1
3 ) (
4     input i_clk,
5     input i_rst,
6     output [23:0] o_bcd_time
7 );
8     reg [3:0] ms_1_r;
9     reg [3:0] ms_10_r;
10    reg [3:0] ms_100_r;
11    reg [3:0] s_1_r;
12    reg [3:0] s_10_r;
13    reg [3:0] m_1_r;
14    assign o_bcd_time = {m_1_r, s_10_r, s_1_r, ms_100_r, ms_10_r, ms_1_r};
15
16    wire sub_ms_wrap;
17    wire ms_1_wrap;
18    wire ms_10_wrap;
19    wire ms_100_wrap;
20    wire s_1_wrap;
21    wire s_10_wrap;
22    wire m_1_wrap;
23
24    generate
25    if (CYCLES_PER_MS > 1) begin : g_sub_ms
26        reg [$clog2(CYCLES_PER_MS)-1:0] sub_ms_r;
27        assign sub_ms_wrap = sub_ms_r == CYCLES_PER_MS-1;
28
29        always @(posedge i_clk, negedge i_rst)
30            if (!i_rst) sub_ms_r <= 0;
31            else sub_ms_r <= sub_ms_wrap ? 0 : (sub_ms_r + 1);
32    end else
33        assign sub_ms_wrap = 1;
34    endgenerate
35    assign ms_1_wrap = ms_1_r == 4'd9 && sub_ms_wrap;
36    assign ms_10_wrap = ms_10_r == 4'd9 && ms_1_wrap;
37    assign ms_100_wrap = ms_100_r == 4'd9 && ms_10_wrap;
38    assign s_1_wrap = s_1_r == 4'd9 && ms_100_wrap;
39    assign s_10_wrap = s_10_r == 4'd5 && s_1_wrap;

```

```

40     assign m_1_wrap = m_1_r == 4'd9 && s_10_wrap;
41
42     always @(posedge i_clk, negedge i_rst) begin
43         if (!i_rst) begin
44             ms_1_r <= 0;
45             ms_10_r <= 0;
46             ms_100_r <= 0;
47             s_1_r <= 0;
48             s_10_r <= 0;
49             m_1_r <= 0;
50         end else begin
51             if (sub_ms_wrap) ms_1_r <= ms_1_wrap ? 0 : (ms_1_r + 1);
52             if (ms_1_wrap) ms_10_r <= ms_10_wrap ? 0 : (ms_10_r + 1);
53             if (ms_10_wrap) ms_100_r <= ms_100_wrap ? 0 : (ms_100_r + 1);
54             if (ms_100_wrap) s_1_r <= s_1_wrap ? 0 : (s_1_r + 1);
55             if (s_1_wrap) s_10_r <= s_10_wrap ? 0 : (s_10_r + 1);
56             if (s_10_wrap) m_1_r <= m_1_wrap ? 0 : (m_1_r + 1);
57         end
58     end
59 endmodule

```